Automating Standard Laptop Orders with Flow Designer

Final Project Report | Platform: ServiceNow (Flow Designer, Service Catalog)

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1. INTRODUCTION

1.1 Project Overview

In many organizations, employees request standard laptops through emails, spreadsheets or phone calls. Each request must be approved, passed to procurement or IT, tracked and confirmed, which causes delays, lost requests and no visibility into status.

This project automates the complete standard laptop ordering process using **ServiceNow Flow Designer**, a low-code workflow tool. An employee submits a request from a **Service Catalog item**. The flow then automatically routes it to the manager for approval, creates fulfilment tasks for IT, sends notifications at each stage and closes the request once the laptop is delivered. No manual hand-offs or coding are needed.

1.2 Purpose

- Replace manual, email-based laptop ordering with a single, standard request form.
- Cut approval and fulfilment time through automatic routing.
- Give requesters and approvers real-time status and an audit trail.
- Enforce company policy by offering only pre-approved standard laptop models.
- Show how low-code automation reduces errors and administrative effort.

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2. IDEATION PHASE

2.1 Problem Statement

Item	Description
I am	An employee who needs a new standard laptop
I'm trying to	Request, get approval for and receive a laptop quickly
But	Requests go through emails and follow-ups, and approvals get delayed or lost
Because	There is no automated, trackable workflow
Which makes me feel	Frustrated, uncertain and unproductive

Problem statement: Standard laptop requests are handled manually, causing delays, missing approvals, inconsistent data and poor visibility for employees, managers and the IT team.

2.2 Empathy Map Canvas

Quadrant	Employee (Requester)

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Says	"Where is my laptop request?" "Who needs to approve this?"
Thinks	"Why does a simple request take so long?"
Does	Sends emails, follows up with the manager and IT, checks status repeatedly
Feels	Frustrated, anxious, dependent on others
Pains	Lost requests, no status updates, repeated follow-ups
Gains	One simple form, automatic approvals, clear tracking, faster delivery

2.3 Brainstorming

Ideas considered by the team:

1. A shared spreadsheet to log requests (rejected: no workflow or audit trail).
2. An email-based approval chain (rejected: hard to track).
3. A custom-coded application (rejected: high effort and maintenance).
4. **A Service Catalog item with Flow Designer automation (selected).** It is low-code, scalable, built into the platform and provides approvals, tasks and notifications out of the box.

Feature ideas prioritized: catalog form with laptop model dropdown, manager approval, auto-created IT task, email notifications, rejection handling, status tracking, and reporting dashboard (future).

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3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Stage	Requester action	Touchpoint	Experience	System opportunity
Awareness	Needs a new laptop	Service Portal	Unsure of process	Clear catalog item
Request	Fills in the form and submits	Catalog item	Simple, quick	Pre-filled user details
Approval	Waits for manager decision	Email / portal	Anxious	Auto approval request and reminders
Fulfilment	IT prepares the laptop	Task / notification	Wants updates	Auto task creation and status
Delivery	Receives the laptop	Notification	Satisfied	Auto-close and feedback

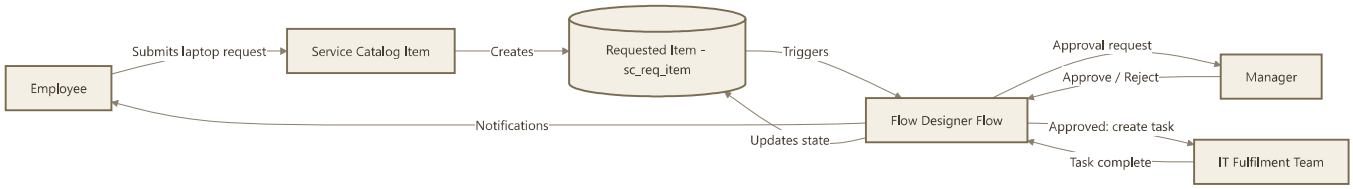
3.2 Solution Requirement

Functional requirements

ID	Requirement
FR-1	Provide a catalog item "Standard Laptop Request" with variables (model, justification, delivery location, required-by date).
FR-2	Trigger a flow automatically when the catalog request is submitted.
FR-3	Send an approval request to the requester's manager.
FR-4	If approved, create a fulfilment (catalog) task for the IT team.
FR-5	If rejected, update the request state and notify the requester.
FR-6	Send email notifications at submission, approval/rejection and completion.
FR-7	Close the request when the fulfilment task is completed.

Non-functional requirements: usability (simple form), reliability (flow executions logged), security (role-based access), scalability (new models added without redesign), auditability (full activity history).

3.3 Data Flow Diagram



3.4 Technology Stack

Layer	Technology
Platform	ServiceNow (Personal Developer Instance)
Automation	Flow Designer (flows, actions, approvals)
Front end	Service Catalog / Service Portal
Data	Platform tables: sc_request, sc_req_item, sc_task, sys_user, sysapproval_approver
Communication	Email notifications
Access control	Roles and groups (catalog users, managers, IT fulfilment group)

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4. PROJECT DESIGN

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4.1 Problem Solution Fit

- **Problem:** manual, slow and untracked laptop ordering.
- **Customer segments:** employees, managers and IT fulfilment staff.
- **Solution fit:** a single catalog form plus an automated flow removes follow-ups, gives clear status, and standardizes models and approvals. It uses a platform the organization already owns, so adoption is simple.

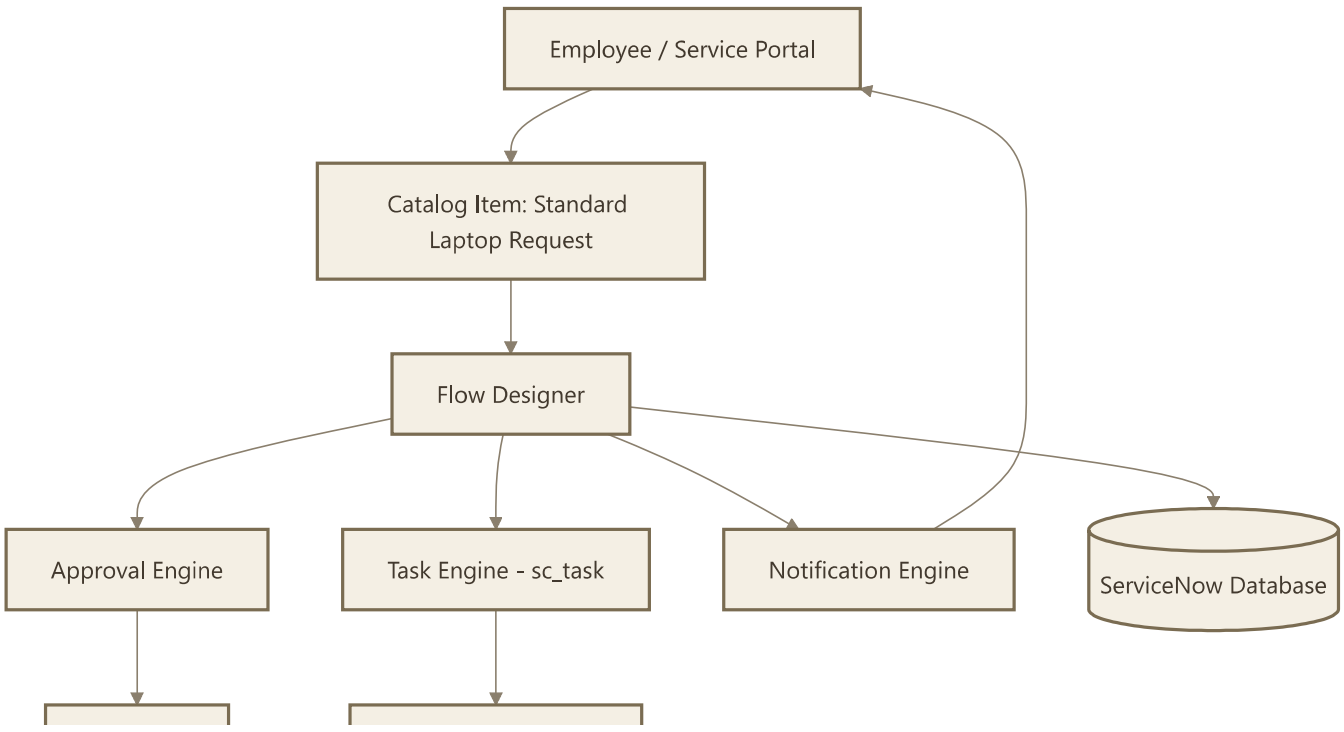
4.2 Proposed Solution

Aspect	Description
Novelty	End-to-end no-code automation of a routine procurement request
Feasibility	Uses standard ServiceNow features, with no custom code required
Business impact	Faster delivery, fewer errors, reduced admin workload
Scalability	The same pattern extends to monitors, phones, software and other catalog items

Flow logic:

1. **Trigger:** Service Catalog, a request item is created for "Standard Laptop Request".
2. **Action:** Ask for approval from the requester's manager.
3. **If approved:** Create a catalog task for the IT group, notify the requester, wait for task completion, then set the item to *Closed Complete*.
4. **If rejected:** Set the item to *Closed Incomplete* and notify the requester with the reason.

4.3 Solution Architecture



Manager | IT Fulfilment Group

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5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Sprint	Activity	Duration	Status
Sprint 1	Requirement gathering, problem statement, empathy map, brainstorming	2 days	Completed
Sprint 2	Set up instance, users, groups and roles; create catalog item and variables	2 days	Completed
Sprint 3	Build flow: trigger, approval, task creation, notifications	3 days	Completed
Sprint 4	Testing, bug fixing and performance validation	2 days	Completed
Sprint 5	Documentation, demo and final report	1 day	Completed

Adjust dates and durations to match your actual schedule.

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6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Test ID	Scenario	Expected result	Actual result	Status
TC-01	Submit a laptop request with all mandatory fields	Request item created; flow triggered	As expected	Pass
TC-02	Submit with a mandatory field blank	Form blocks submission	As expected	Pass
TC-03	Manager approves	IT task created; requester notified	As expected	Pass
TC-04	Manager rejects	Item closed incomplete; rejection email sent	As expected	Pass
TC-05	IT completes the task	Request item closed complete; notification sent	As expected	Pass
TC-06	Check flow execution history	All steps logged without errors	As	Pass

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TC-07	Multiple requests submitted back to back	Each runs its own flow instance	As expected	Pass

Performance observations: the flow starts within seconds of submission, and the end-to-end automation overhead (excluding human approval and delivery time) is negligible. *Replace with your measured values.*

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7. RESULTS

7.1 Output Screenshots

Insert your screenshots below.

1. Service Catalog item "Standard Laptop Request" form
2. Catalog item variables configuration
3. Flow Designer: trigger and flow canvas
4. Approval request in the manager's view
5. Generated catalog task for the IT team
6. Email notifications received
7. Flow execution history (completed run)
8. Request item in *Closed Complete* state

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8. ADVANTAGES & DISADVANTAGES

Advantages

- Faster request-to-delivery cycle through automatic routing
- Standardized models and policy compliance
- Complete audit trail and real-time status visibility
- Low-code: easy to build, modify and maintain
- Fewer manual errors and less follow-up effort
- Reusable for other catalog items

Disadvantages

- Depends on ServiceNow licensing and platform availability
- Limited to predefined standard models; special requests need a separate process
- Approval speed still depends on the manager responding
- Initial setup needs correct user, group and manager data
- Not integrated with inventory or procurement systems yet

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9. CONCLUSION

The project automated the standard laptop ordering process using ServiceNow Flow Designer. By combining a catalog item with an approval-driven flow, the solution removes manual hand-offs, speeds up fulfilment, improves transparency and ensures every request follows company policy. It demonstrates that low-code automation can deliver measurable efficiency gains with minimal development effort and can be extended to many other service requests.

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10. FUTURE SCOPE

- Integrate with inventory and asset management to check stock and auto-create asset records.
- Connect to procurement or vendor systems for out-of-stock orders.
- Add multi-level approvals based on cost or department.
- Add reminders and escalations for pending approvals.
- Build a dashboard with order volume and cycle-time reports.
- Provide a mobile/virtual-agent (chatbot) request channel.
- Extend to other hardware and software requests.

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11. APPENDIX

Source Code (if any)

No custom code was required. The solution is configured using Flow Designer and Service Catalog. Update set / flow export: *[add file or description here]*

Dataset Link

Not applicable (uses ServiceNow platform tables). Sample users and catalog data: *[add link if any]*

GitHub & Project Demo Link

- GitHub: Streamlining-IT-Procurement-Automating-Standard-Laptop-Orders-with-Flow-Designer
- Demo video: *[add link]*